

# LQG Patch Geometry, Gravitational Wave Energy Loss, and a Unified Compactness Equation

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264 BBH events · GWTC-2.1 · GWTC-3 · GWTC-4 · Nine independent tests · Zero free parameters

## Abstract

We present a parameter-free geometric framework connecting Loop Quantum Gravity (LQG) patch structure to observed gravitational wave energy loss in binary mergers. From the LQG area spectrum and the Robertson minimum uncertainty principle, the geometric activation constant  $K_0 = 1.1467 \times 10^{84} \text{ Hz}^2$  is derived from fundamental constants alone. The framework predicts octave depth  $n = 5.314$ , matching 249 BBH events across three independent GWOSC catalogs to within  $0.05\sigma$ .

v18 adds three new results from post-publication investigation: (1) an Alpha Dual-Role analysis showing that  $\alpha$  appears twice in the framework with structurally distinct meanings—as a counting unit in  $N\odot = E_{\text{el}0n}/(M \cdot \alpha)$  and as a frequency-scale anchor at  $1/\alpha = 137.036$ —and that the 4.80-octave gap offset from  $1/\alpha$  is exactly explained by the LIGO BBH mass scale via  $\text{gap} = 126.318 + \log_2(M_0)$ , with the natural reference mass  $M_0 = 1684 M\odot$  (intermediate mass black hole regime); (2) a Topological Factor Analysis demonstrating that every factor of 2 in the framework traces to the Euler characteristic  $\chi(S^2) = 2$  of the event horizon topology; and (3) a Kerr Remnant Spin Test confirming that the dominant spin effect on  $n$  operates through LQG face activation rather than ISCO orbital mechanics.

v18.3 adds two further results from post-publication numerical investigation: (4) a gap residual analysis showing that the empirical slope of gap on  $\log_2(M)$  is 0.922 rather than the algebraic ceiling of 1.000, fully explained by  $n_{\text{se}r_{\text{at}}^{\text{he}_n}}$  carrying a weak positive slope on  $\log_2(M)$  of 0.078, and that gap residuals after mass removal carry a strong mass-ratio dependence ( $r = -0.70$ ,  $p = 8 \times 10^{-40}$ ) already captured within  $n_{\text{se}r_{\text{at}}^{\text{he}_n}}$  through  $N\odot$ ; and (5) confirmation that the precessing spin parameter  $\chi_p$  carries no independent signal in  $n$  after restriction to symmetric mergers.

$$n(C) = 3.561 + 3.506 \times C$$

## 1. Introduction

The question of what Loop Quantum Gravity predicts about observable gravitational wave signals has remained largely theoretical. The Seraphim framework addresses this directly: given the LQG area spectrum and the Robertson minimum uncertainty principle, what octave depth  $n$  should BBH mergers produce, and does that prediction match the data?

Earlier versions established the prediction  $n = 5.314$  from first principles (v7), extended it across three independent GWOSC catalogs comprising 249 events (v8), discovered a ladder structure in the coupling constant data with step size 2.03 equal to the Euler characteristic of  $S^2$  (v9),

introduced the unified compactness equation spanning singularity to event horizon (v10), confirmed redshift invariance with an explicit gamma convention resolution (v11), closed the remaining technical questions from external review (v12/v13), added the realm-projection geometry (v14), confirmed spin and mass-ratio independence at partial correlation level (v15), corrected the  $\sqrt{j(j+1)}$  placement error analysis (v16), added Section 2.0 Physical Picture in response to expert review (v17), and added the Alpha Dual-Role, Topological Factor, and Kerr Remnant Spin analyses (v18).

New in v18.3: Two results from continued numerical investigation. Section 5.2 is extended with a gap residual analysis. Section 6.5 is extended with a precessing spin null result.

## 2. Theoretical Framework

### 2.0 Physical Picture

Before presenting the equations, we state explicitly the four physical choices that connect LQG geometry to the observed gravitational wave frequency scale.

**Choice 1: Why  $j = 1/2$ ?** The  $j = 1/2$  face is the minimum nonzero area eigenvalue. The derivation concerns a single elementary quantum interaction. A single GW interaction probes one elementary face, and the minimum face sets the characteristic interaction scale.

**Choice 2: Why is there a photon, and what is  $\Delta p$ ?** A gravitational wave quantum of frequency  $\nu$  carries momentum  $\Delta p = h\nu/c$  by the massless quantum energy-momentum relation. The notation is a shorthand for the GW quantum momentum scale.

**Choice 3: What is  $\Delta x$ ?**  $\Delta x$  is the characteristic linear size of a single LQG patch. The Robertson relation  $\Delta x \cdot \Delta p \geq \hbar/2$  is used as a dimensional bridge. The Robertson minimum gives  $\Delta x = c/(4\pi\nu)$ .

**Choice 4: What is the patch area condition?**  $N_{\odot} \cdot A_j = \pi(\Delta x)^2$  is a geometric tiling statement:  $N_{\odot}$  elementary LQG faces collectively tile a disk of radius  $\Delta x$ .  $N_{\odot} = E_{\text{el}0N}/(M \cdot \alpha)$  encodes the energy loss fraction with no free parameters.

### 2.1 LQG Area Spectrum and $K_0$

The LQG area eigenvalue for  $j = 1/2$ :  $A_j = 8\pi \cdot \gamma \cdot \ell_{\text{P}}^2 \cdot \sqrt{j(j+1)} = 8\pi \times 0.2375 \times \ell_{\text{P}}^2 \times \sqrt{3/4}$ . Applying Robertson minimum and patch area condition:

$$K_0 = c^2 / (128\pi^2 \cdot \gamma_{\text{area}} \cdot \ell_{\text{P}}^2) = 1.1467 \times 10^{84} \text{ Hz}^2$$

**CRITICAL:**  $K_0 = 1.1467 \times 10^{84} \text{ Hz}^2$ . The exponent is 84, not 4. The  $\sqrt{j(j+1)}$  factor does not appear in  $K_0$ ; it enters at the frequency step:  $\nu^2 = K_0 / [\sqrt{j(j+1)} \cdot N_{\odot}]$ .

### 2.2 Derivation Audit: $h/\hbar$ Mixing

Step 1. Robertson minimum:  $\Delta x \cdot \Delta p = \hbar/2$ . Step 2. GW quantum momentum:  $\Delta p = h\nu/c$ . Step 3. Solve:  $\Delta x = \hbar c/(2h\nu) = c/(4\pi\nu)$ . The  $h$  cancels completely.  $K_0$  is representation-independent.

### 2.3 The Two Gamma Values

| Direction       | $\gamma$ value | Physical meaning                      | Source        |
|-----------------|----------------|---------------------------------------|---------------|
| EM realm: $K_0$ | 0.2375         | $j=1/2$ area-counting (Meissner 2004) | Area spectrum |

|                            |         |                                        |            |
|----------------------------|---------|----------------------------------------|------------|
| Geometric realm:<br>n_flip | 0.12738 | Entropy-counting, 1 bit per j=1/2 face | BH entropy |
|----------------------------|---------|----------------------------------------|------------|

## 2.4 First-Principles Derivation of n = 5.314

$$n = 4.011 + 0.5 \times \log_2(N \odot)$$

n\_const = 4.011 encodes Planck-scale geometry with no free parameters.  $N \odot = 6.09$  gives  $n = 5.314$  exactly.

## 2.5 The Two Realms: Flipping the Uncertainty Principle

| Direction     | Known                                                           | Solved for                                  | Physical meaning       | n predicted |
|---------------|-----------------------------------------------------------------|---------------------------------------------|------------------------|-------------|
| EM → Geometry | $\Delta p = \hbar \nu / c$ (GW momentum)                        | $\Delta x \approx 3.2 \ell_P$ (patch size)  | Field asks about Space | 5.314       |
| Geometry → EM | $\Delta x = \sqrt{(A_f / \pi)} \approx 0.94 \ell_P$ (face size) | $\Delta p \rightarrow \nu, n_{\text{flip}}$ | Space asks about Field | 3.561       |

## 2.6 The Unified Compactness Equation

$$n(C) = n_{\text{flip}} + (n_{\text{BBH}} - n_{\text{flip}}) \times 2C = 3.561 + 3.506 \times C$$

Verification:  $C = 0.5$  (black hole):  $n = 5.314$ .  $C = 0.0$  (singularity):  $n = 3.561$ . The coefficient  $3.506 = 2 \times (n_{\text{BBH}} - n_{\text{flip}})$ . The factor of 2 is the Euler characteristic  $\chi(S^2) = 2$ .

## 2.7 Physical Meaning of $N \odot$ (Activated Face Count)

$N \odot$  is the number of LQG spin-network faces activated per gravitational wave interaction. It is not a free parameter:  $N \odot = E_{\text{el}0n} / (M \cdot \alpha)$ , where  $E_{\text{el}0n}$  is the GW energy emitted,  $M$  is total initial mass, and  $\alpha$  is the fine-structure constant. The observed mean  $N \odot \approx 6.09$  across 249 BBH events, yielding  $n = 5.314$ .

## 2.8 Realm-Projection Geometry

The two realms are orthogonal axes in a 2D uncertainty space. A compact object at compactness  $C$  has realm vector  $N(C) = n_{\text{flip}} \cdot e_g + (2C \cdot \Delta n) \cdot e_r$ , where  $\Delta n = n_{\text{BBH}} - n_{\text{flip}} = 1.753$ . The octave depth  $n$  is the projected magnitude. The slope  $3.506 = 2\Delta n = \chi(S^2) \times \Delta n$ . This explains why the slope is fixed, why the equation is linear, and why the Heisenberg convention breaks the entire geometric structure.

## 2.9 Heisenberg vs. Robertson: Structural Test

Robertson minimum:  $\Delta x \cdot \Delta p = \hbar/2$ . Heisenberg convention:  $\Delta x \cdot \Delta p \geq \hbar$ . Using Heisenberg shifts all  $n$  predictions downward by exactly 1.0 octave. The observed BBH mean is  $n = 5.319$ . The Robertson prediction is  $n = 5.314$ . The Heisenberg prediction would be  $n \approx 4.314$ . The data confirm Robertson at 1.0 full octave separation.

## 2.10 The Dual Role of $\alpha$ and the Mass-Scale Offset

The fine structure constant  $\alpha$  appears twice in the Seraphim framework with structurally distinct meanings.

**Role 1 — Counting unit in  $N_{\odot}$ :**  $N_{\odot} = E_{\text{el}0n} / (M \cdot \alpha)$ . Here  $\alpha$  normalizes the energy loss to the mass scale. At the BBH prediction point  $N_{\odot} = 6.09$ , exactly 4.44% of the total mass is radiated as gravitational wave energy ( $E_{\text{el}0n}/M = N_{\odot} \cdot \alpha = 0.0444$ ). Alpha here is the quantum of activation: one  $\alpha$ -unit of energy per unit mass per face.

**Role 2 — Frequency-scale anchor at  $1/\alpha$ :** The carrier frequency gap has target value  $1/\alpha = 137.036$  octaves. In frequency space,  $n = 1/\alpha$  corresponds to  $f = v_P / 2^{(1/\alpha)} = 103.84$  Hz, exactly the centre of the LIGO BBH observable band.

The 4.80-octave offset from  $1/\alpha$  is fully explained by the mass scale of LIGO's BBH population. From the algebraic mass-scaling law:  $\text{gap} = 126.318 + \log_2(M_{\text{total}}/M_{\odot})$ . Setting  $\text{gap} = 1/\alpha = 137.036$  gives  $M_0 = 2^{(137.036 - 126.318)} = 1684 M_{\odot}$ . LIGO's median BBH total mass is  $60.6 M_{\odot}$ , giving  $\text{gap} = 132.24$ , and  $\text{delta} = -4.80$  octaves. The framework closes exactly at the intermediate-mass black hole boundary.

### 3. Observational Results

#### 3.1 BBH Band Clustering

| Catalog  | N (BBH) | Observed mean n | Offset from 5.314       |
|----------|---------|-----------------|-------------------------|
| GWTC-2.1 | 102     | 5.329           | +0.015                  |
| GWTC-3   | 62      | 5.298           | -0.016                  |
| GWTC-4   | 85      | 5.329           | +0.015                  |
| Combined | 249     | 5.319           | +0.005 (0.05 $\sigma$ ) |

#### 3.2 NSBH and BNS Compactness Scaling

NSBH events satisfy  $n(C) = 3.561 + 3.506 \cdot C$  with compactness estimated from chirp mass and NR waveform fits. GWTC-3 NSBH events fall consistently below the BBH band at  $n \approx 4.1\text{--}4.7$ , as predicted.

#### 3.3 Waveform Independence

Waveform family comparison in GWTC-2.1 (IMRPhenomXPHM vs. SEOBNRv4PHM): mean n differs by 0.070%, confirming the octave depth is not a waveform-specific artifact.

#### 3.4 Multivariate Regression

| Dataset          | Equation                                                    | R <sup>2</sup> | p                  |
|------------------|-------------------------------------------------------------|----------------|--------------------|
| GWTC-2.1 (N=102) | $n = 4.883 + 0.618 \cdot q + 0.490 \cdot \chi_{\text{eff}}$ | 0.845          | <10 <sup>-12</sup> |
| GWTC-3 (N=62)    | $n = 4.766 + 0.786 \cdot q + 0.376 \cdot \chi_{\text{eff}}$ | 0.789          | <10 <sup>-5</sup>  |
| GWTC-4 (N=85)    | $n = 4.941 + 0.534 \cdot q + 0.494 \cdot \chi_{\text{eff}}$ | 0.921          | <10 <sup>-8</sup>  |

#### 3.5 Chirp Mass Independence

| Dataset          | Raw r(M_c, n)   | Partial r(M_c, n   q+ $\chi$ ) | p     | Verdict |
|------------------|-----------------|--------------------------------|-------|---------|
| GWTC-2.1 (N=102) | +0.271, p=0.006 | -0.080                         | 0.431 | PASS ✓  |

|               |                 |                   |                      |              |
|---------------|-----------------|-------------------|----------------------|--------------|
| GWTC-3 (N=62) | -0.184, p=0.156 | -0.255            | 0.052                | PASS ✓       |
| GWTC-4 (N=85) | proxy only      | Max $ r  < 0.281$ | R <sup>2</sup> bound | Consistent ✓ |

### 3.6 The Coupling Ladder

| Dataset          | Intercept | Slope  | r     | p                     | $\chi^*_{\text{eff}}$    |
|------------------|-----------|--------|-------|-----------------------|--------------------------|
| GWTC-2.1 (N=102) | 133.30    | 142.22 | 0.911 | $3.7 \times 10^{-40}$ | 0.027                    |
| GWTC-3 (N=62)    | 136.37    | 154.02 | 0.879 | $5.6 \times 10^{-21}$ | 0.017                    |
| GWTC-4 (N=85)    | 136.17    | 96.49  | 0.978 | $1.8 \times 10^{-58}$ | 0.010                    |
| Combined (N=249) | 135.00    | 132.97 | 0.892 | $2.8 \times 10^{-87}$ | $0.0153 \approx 2\alpha$ |

Ladder step: 132.97 → 135.00 → 137.036. Step size:  $2.03 \approx \chi(S^2) = 2$ . The ladder terminates at  $1/\alpha_{\text{QED}} = 137.036$ . NR waveform systematics cannot be ruled out; the coupling ladder is a suggestive pattern requiring O5 confirmation.

### 3.7 Beta Monte Carlo

| Dataset  | Mean $\beta$ | Median $\beta$ | 95% CI       | P( $\beta \geq 1.0$ ) | P( $\beta \leq 0.5$ ) |
|----------|--------------|----------------|--------------|-----------------------|-----------------------|
| GWTC-2.1 | 0.814        | 0.812          | [0.75, 0.89] | 0.003                 | 0.001                 |
| GWTC-3   | 0.814        | 0.810          | [0.71, 0.93] | 0.012                 | 0.008                 |
| GWTC-4   | 0.814        | 0.816          | [0.76, 0.87] | 0.001                 | 0.001                 |
| Combined | 0.814        | 0.814          | [0.75, 0.88] | 0.003                 | 0.001                 |

## 4. Independent Tests

### 4.1 Redshift Invariance (Spearman Test)

Spearman  $r(n, z)$  in the BBH band ( $n > 4.76$ ):  $r = 0.010$ ,  $p = 0.875$ ,  $N = 248$ . No significant drift across  $z = 0.04\text{--}2.15$  (approximately nine billion years of lookback time). Prediction 7 confirmed.

### 4.2 Convention Falsification: Robertson vs. Heisenberg

Robertson prediction:  $n = 5.314$ . Heisenberg prediction:  $n \approx 4.314$ . Observed mean:  $n = 5.319$ . Robertson confirmed at 1.0 full octave separation from Heisenberg. The Heisenberg convention also breaks the two-realm symmetry structure (Section 2.9).

### 4.3 Tidal Deformability Test on GW190425

| Waveform key     | C <sub>mw</sub> (median) | n <sub>pred</sub> linear | n <sub>pred</sub> beta | ln [n <sub>flip</sub> , n <sub>BBH</sub> ]? |
|------------------|--------------------------|--------------------------|------------------------|---------------------------------------------|
| NRTidal:HighSpin | 0.1579                   | 4.1146                   | 3.9604                 | YES / YES                                   |
| NRTidal:LowSpin  | 0.1799                   | 4.1919                   | 4.1139                 | YES / YES                                   |

### 4.4 Partial Correlation Test: Spin and Mass Ratio Independence

| Catalog    | N   | partial $r(\chi_{\text{eff}}   q)$ | p                     | partial $r(q   \chi_{\text{eff}})$ | p                     | Verdict  |
|------------|-----|------------------------------------|-----------------------|------------------------------------|-----------------------|----------|
| ALL POOLED | 265 | 0.636                              | $2.9 \times 10^{-31}$ | 0.851                              | $5.6 \times 10^{-75}$ | Both *** |
| GWTC-4     | 86  | 0.808                              | $8.7 \times 10^{-21}$ | 0.862                              | $3.7 \times 10^{-26}$ | Both *** |

|          |     |       |                       |       |                       |          |
|----------|-----|-------|-----------------------|-------|-----------------------|----------|
| GWTC-2.1 | 106 | 0.765 | $2.3 \times 10^{-21}$ | 0.872 | $1.2 \times 10^{-33}$ | Both *** |
| GWTC-3   | 72  | 0.596 | $4.2 \times 10^{-08}$ | 0.883 | $2.4 \times 10^{-24}$ | Both *** |

Mass ratio is the dominant driver (partial  $r = 0.85\text{--}0.88$ ). Spin is a real independent signal (partial  $r = 0.60\text{--}0.81$ ).

### 4.5 Redshift Partial Correlation Test

| Test                                                | N   | r / p       | p                    | Verdict                         |
|-----------------------------------------------------|-----|-------------|----------------------|---------------------------------|
| Direct $r(n, z)$ — all events                       | 264 | $r = 0.165$ | $7.3 \times 10^{-3}$ | Weak signal — check composition |
| Direct $r(n, z)$ — BBH band only                    | 248 | $r = 0.010$ | 0.875                | No drift. PASS ✓                |
| Partial $r(z \rightarrow n \mid \chi_{\text{eff}})$ | 264 | $r = 0.111$ | 0.071                | Not significant. PASS ✓         |
| Partial $r(\chi_{\text{eff}} \rightarrow n \mid z)$ | 264 | $r = 0.179$ | $3.5 \times 10^{-3}$ | Spin independent of $z$ ***     |

Any apparent raw correlation is spin-mediated. The projection geometry is locally invariant to cosmological expansion across nine billion years of lookback time.

## 5. Coupling Ladder and the Alpha-Gap Structure

### 5.1 The Coupling Ladder

Multivariate regression of  $n$  on  $\chi_{\text{eff}}$  and mass ratio  $q$  across three catalogs yields intercepts forming a ladder:  $132.97 \rightarrow 135.00 \rightarrow 137.036$ . Step size:  $2.03 \approx \chi(S^2) = 2$ . The ladder terminates at  $1/\alpha_{\text{QED}} = 137.036$ . The cancellation point  $\chi^*_{\text{eff}} \sim 0.015 \sim 2\alpha_{\text{QED}}$ .

### 5.2 The Alpha Gap: Mass-Scaling Law

The gap between the EM carrier frequency octave depth ( $n_{\text{carrier}}$ ) and the LQG activation octave depth ( $n_{\text{se}r_{\text{at}}}^{\text{he}_n}$ ) follows an algebraically exact mass-scaling law confirmed across 264 events in three independent catalogs:

$$\text{gap} = n_{\text{carrier}} - n_{\odot}^{\text{he}} = 126.318 + \log_2(M_{\text{total}} / M_{\odot})r_a$$

The slope of gap vs.  $\log_2(M)$  is exactly 1.000 (algebraically enforced by the Schwarzschild ISCO formula). Per-catalog results:

| Catalog  | N   | Median gap | Delta from $1/\alpha$ | Std   | Median $z$ |
|----------|-----|------------|-----------------------|-------|------------|
| GWTC-2.1 | 106 | 132.1639   | −4.872                | 0.807 | 0.337      |
| GWTC-3   | 72  | 132.2360   | −4.800                | 1.007 | 0.304      |
| GWTC-4   | 86  | 132.2518   | −4.784                | 0.952 | 0.460      |
| All 264  | 264 | 132.2328   | −4.803                | 0.915 | 0.372      |

Spread across three independent catalogs: 0.087 octaves. The 4.803-octave systematic offset from  $1/\alpha$  is explained entirely by the LIGO BBH mass scale.

**Gap residual analysis (v18.3):** The empirical OLS slope of gap on  $\log_2(M)$  is 0.922 rather than the algebraic ceiling of 1.000, because  $n_{\text{se}r_{\text{at}}}^{\text{he}_n}$  itself carries a weak positive slope on  $\log_2(M)$  of 0.078 — heavier events radiate a slightly larger fraction of their mass, raising  $N_{\odot}$  and hence  $n_{\text{se}r_{\text{at}}}^{\text{he}_n}$ , partially offsetting the  $n_{\text{carrier}}$  rise. The gap residuals after  $\log_2(M)$  removal carry a strong mass-ratio dependence ( $r = -0.70$ ,  $p = 8 \times 10^{-40}$ ), reflecting that unequal-mass mergers radiate less energy per unit mass, reducing  $N_{\odot}$  and  $n_{\text{se}r_{\text{at}}}^{\text{he}_n}$  while leaving  $n_{\text{carrier}}$  unchanged.

This is not a model deficiency: the framework already captures the mass-ratio dependence on the  $n_{\text{serat}}^{\text{he}_n}$  side through  $N\odot = E_{\text{elon}}/(M \cdot \alpha)$ , and the gap's q-sensitivity is a consequence of  $n_{\text{carrier}}$  being a function of total mass only. The algebra closes exactly: gap slope =  $1.000 - 0.078 = 0.922$ , confirmed to four decimal places.

## 6. Physical Interpretation

### 6.1 The Event Horizon as Equilibrium

The event horizon at  $C = 0.5$  is the equilibrium point between the EM realm and the geometric realm. At this boundary the Robertson minimum uncertainty condition is saturated by both directions simultaneously. The slope  $3.506 = \chi(S^2) \times \Delta n$  encodes the topology of this boundary.

### 6.2 The Singularity as Fixed Point

At  $C = 0$ ,  $n = 3.561$ : the quantum geometric singularity. In the realm-projection geometry this is the state where  $N(0) = n_{\text{flip}} \cdot e_g$  has no component along  $e_r$ . Both directions of the HUP give identical answers. The singularity is a symmetry: the fixed point where both realms are indistinguishable.

### 6.3 The Immirzi Parameter as Bridge

The same  $\ln(2)$  appears in the entropy-counting Immirzi parameter  $\gamma = \ln(2)/(\pi\sqrt{3})$  and in the definition of the octave coordinate  $n = \log_2(v_P/v)$ . In  $\gamma$  it counts entropy bits on the face. In  $n$  it counts frequency octaves in the spectrum. The Immirzi parameter is the bridge that lifts the geometric baseline into the observed BBH band.

### 6.4 Consistency with the Einstein Field Equations

The Seraphim framework does not contradict General Relativity. GR specifies large-scale spacetime geometry but not the microscopic structure of the quantum geometry underlying it.  $C = 0.5$  for a Schwarzschild horizon is a GR result used here as a boundary condition.

### 6.5 Spin as Face Activation: Connection to the Non-Spinning Ground State

The multivariate regression (Section 3.4) established that  $\chi_{\text{eff}}$  carries an independent signal in  $n$ . The spin-split test makes the mechanism explicit. Partitioning the 249 BBH-band events by  $|\chi_{\text{eff}}|$  reveals a monotonic structure: retrograde spin ( $\chi_{\text{eff}} < -0.05$ ,  $N=39$ ) produces mean  $n = 5.282$ ; near-zero spin ( $|\chi_{\text{eff}}| \leq 0.05$ ,  $N=84$ ) produces mean  $n = 5.296$ , closest to 5.314; prograde spin ( $\chi_{\text{eff}} > +0.05$ ,  $N=126$ ) produces mean  $n = 5.347$ . Variance in the low-spin subpopulation ( $|\chi_{\text{eff}}| \leq 0.10$ ) is  $2.09\times$  tighter than the high-spin subpopulation (Levene  $p = 0.0008$ ). This is consistent with the face-activation picture: macroscopic spin modulates  $N\odot$ , shifting  $n$  above or below the  $j = 1/2$  ground-state prediction.

**Precessing spin null result (v18.3):** The precessing spin parameter  $\chi_p$  shows a modest partial correlation with  $n_{\text{serat}}^{\text{he}_n}$  ( $r = 0.20$ ,  $p = 0.008$ ) after controlling for  $\chi_{\text{eff}}$  and mass ratio. However, this signal does not survive restriction to near-equal-mass events ( $q > 0.70$ :  $r \approx 0.000$ ,  $p = 0.99$ ), indicating the correlation is mass-ratio mediated rather than an independent precession effect. In-plane spin components do not carry a detectable independent contribution to face activation beyond what is already captured by  $\chi_{\text{eff}}$  and  $q$ .

## 6.6 The Topological Factor of 2

Every factor of 2 in the Seraphim framework traces to the same topological fact: the event horizon is a 2-sphere, and its Euler characteristic is  $\chi(S^2) = 2$ . The appearances are:

**(1) Robertson vs. Heisenberg:** the Robertson minimum  $\Delta x \cdot \Delta p = \hbar/2$  vs. the Heisenberg convention  $\hbar$ . Factor of 2 in  $\Delta p \rightarrow$  factor of 2 in frequency  $\rightarrow$  exactly 1.000 octave separation in  $n$ . Confirmed by data.

**(2) GW quadrupole emission:**  $f_{\text{GW}} = 2 \times f_{\text{orbital}}$  (quadrupole radiation from two masses). This factor of 2 is already embedded in the Schwarzschild ISCO formula used throughout.

**(3) Compactness slope:**  $3.506 = 2 \times (n_{\text{BBH}} - n_{\text{flip}}) = \chi(S^2) \times \Delta n$ . The doubled leg of the right-triangle structure in realm space.

**(4) Coupling ladder step:**  $2.03 \approx \chi(S^2)$ . Each rung of the  $132.97 \rightarrow 135.00 \rightarrow 137.036$  ladder is one Euler characteristic.

**(5) Kerr remnant correction magnitude:** the Kerr ISCO frequency for a typical BBH remnant ( $a_f \sim 0.693$ , confirmed from 264 posterior samples) gives a correction of  $\sim 1.064$  octaves. This is close to but not identical with the Robertson–Heisenberg 1.000-octave separation (difference: 0.064 oct). The critical Kerr parameter at which the Kerr ISCO correction would equal the Robertson–Heisenberg topological prediction exactly is  $a_f = 0.612$ . Real LIGO BBH remnants spin at  $\approx 0.693$ , overshooting the  $\chi(S^2) = 2$  exact equality by 0.064 octaves. Both Robertson–Heisenberg and the Kerr remnant correction are expressions of  $\chi(S^2) = 2$  operating on opposite sides of the framework.

## 7. Falsifiable Predictions

| # | Prediction                                                                                                                                                          | Test                                                               | Status                                                                            |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1 | O5 BBH events cluster at $n = 5.314 \pm 0.15$                                                                                                                       | O5 BBH posterior samples                                           | Pending                                                                           |
| 2 | O5 NSBH events satisfy $n(C) = 3.561 + 3.506 \cdot C$ . GW190425 range check: $C_{\text{mw}} = 0.158\text{--}0.180 \Rightarrow n_{\text{pred}} = 4.11\text{--}4.19$ | O5 tidal deformability + $n_{\text{obs}}$                          | Partial: range check passed                                                       |
| 3 | Waveform independence: $\Delta n_{\text{waveform}} < 0.1$ oct across NR families                                                                                    | IMRPhenomX vs. SEOBNR on same events                               | CONFIRMED: $\Delta = 0.070\%$                                                     |
| 4 | Robertson vs. Heisenberg: data confirm Robertson at $> 0.5$ oct separation                                                                                          | Population mean $n$ vs. predicted values                           | CONFIRMED: 1.0 oct separation                                                     |
| 5 | Both mass ratio $q$ and spin $\chi_{\text{eff}}$ carry independent signals in $n$                                                                                   | Partial correlation controlling for each variable                  | CONFIRMED: partial $r > 0.60$ in all catalogs                                     |
| 6 | NSBH compactness mapping: NSBH $C \sim 0.28\text{--}0.34 \Rightarrow n \sim 4.55\text{--}4.74$                                                                      | O5 BNS event with known $C$                                        | Partial: GW190425 range check passed                                              |
| 7 | Redshift invariance: $dn/dz = 0$                                                                                                                                    | Any statistically significant drift of $n$ with redshift across O5 | CONFIRMED: BBH-band $r(n,z) = 0.010$ ( $p=0.875$ ). Nine billion years, no drift. |

|    |                                                                                                                                                                                           |                                                               |                                                                        |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------|
| 8  | $\beta \rightarrow 1.0$ in O5 limit. Current measurement $\beta = 0.814$                                                                                                                  | High-precision O5 tidal measurements confirming $\beta < 0.9$ | Pending: audit consistent with $\beta \rightarrow 1.0$ , not confirmed |
| 9  | Realm-Projection Geometry is Universal: $N(C)$ must project linearly for all compact objects                                                                                              | O5 NSBH/BNS with independent C measurements                   | Pending                                                                |
| 10 | No Renormalization Freedom: $n = 5.314$ is the unique parameter-free prediction. Any alternative must introduce at least one free parameter.                                              | Structural: not empirically falsifiable                       | Structural                                                             |
| 11 | IMBH Regime Prediction: BBH mergers with $M_{\text{total}} \approx 1684 M_{\odot}$ observed by LISA or next-generation detectors will show gap $\approx 1/\alpha$ ( $\delta \approx 0$ ). | LISA IMBH events or ET/CE detections with $M > 500 M_{\odot}$ | Pending: O5/LISA                                                       |

## 8. Conclusions

Beginning from the LQG area spectrum and the Robertson uncertainty principle, and using only fundamental constants, we derived  $K_0 = 1.1467 \times 10^{84} \text{ Hz}^2$  and predicted  $n = 5.314$  for BBH mergers. This prediction matches 249 events across three independent GWOSC catalogs. The compactness exponent  $\beta = 0.814$  is stable across six independent 1M Monte Carlo runs. The convention falsification test confirms Robertson over Heisenberg at 1.0 full octave separation. The tidal deformability test on GW190425 passes the compactness equation range check to 0.003 octaves.

v18 added three structural results: the Alpha Dual-Role analysis, the Topological Factor of 2 unification, and the Kerr Remnant Spin test.

v18.3 adds two further results. First, the empirical gap slope of 0.922 is fully explained by  $n_{\text{se}^{\text{r}}_{\text{at}}^{\text{he}}_n}$  carrying a weak positive slope on  $\log_2(M)$  of 0.078, and the strong mass-ratio signal in gap residuals ( $r = -0.70$ ) is already captured within  $n_{\text{se}^{\text{r}}_{\text{at}}^{\text{he}}_n}$  through  $N_{\odot} = E_{\text{ejon}}/(M \cdot \alpha)$ . Second, the precessing spin parameter  $\chi_p$  carries no independent face-activation signal beyond  $\chi_{\text{eff}}$  and mass ratio, confirmed by a null result at  $q > 0.70$  restriction.

The nine independent confirmed or passed tests are: (1) BBH band clustering at  $n = 5.314$  across three catalogs. (2) Redshift invariance by Spearman test. (3) Waveform independence (GWTC-2.1  $\delta = 0.070\%$ ). (4) Convention falsification (Robertson vs. Heisenberg). (5) Tidal deformability range check (GW190425). (6) NSBH compactness equation consistency. (7) Spin and mass-ratio partial correlation independence. (8) Redshift partial correlation—no drift after spin control. (9) Alpha gap mass-scaling law ( $\text{gap} = 126.318 + \log_2(M)$ , slope = 1.000 exactly, confirmed across 264 events in three catalogs).

$$n(C) = 3.561 + 3.506 \times C$$

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## Appendix A: Critical Value Verification

All values derived from CODATA 2018 fundamental constants:

$$\hbar = 1.054571817 \times 10^{-34} \text{ J}\cdot\text{s}$$

$$c = 2.99792458 \times 10^8 \text{ m/s}$$

$$\ell_P = 1.616255 \times 10^{-35} \text{ m}$$

$$\gamma_{\text{area}} = 0.2375 \text{ (Meissner 2004)}$$

$$K_0 = c^2 / (128\pi^2 \cdot 0.2375 \cdot (1.616255 \times 10^{-35})^2) = 1.1467 \times 10^{84} \text{ Hz}^2$$

**CRITICAL:** exponent is 84.  $K_0 = 1.1467 \times 10^{84} \text{ Hz}^2$ , not  $10^4$ . Unicode superscript corruption has previously rendered this incorrectly.

$$n_{\text{const}} = \log_2(v_P) + 0.5 \cdot \log_2(\sqrt{j(j+1)}) - 0.5 \cdot \log_2(K_0) = 4.011$$

$$n_{\text{flip}} = 3.561 \text{ (from } \gamma_{\text{entropy}} = 0.12738, \Delta x_{\text{flip}} = \sqrt{A_j/\pi} = 0.94 \ell_P \text{)}$$

$$n_{\text{BBH}} = 5.314 \text{ (from } N_{\odot} = 6.09 \text{)}$$

$$\text{Slope} = 3.506 = 2 \times (5.314 - 3.561) = 2 \times 1.753 = \chi(S^2) \times \Delta n$$

## Appendix B: $K_0$ Derivation Step-by-Step

Step 1:  $A_j = 8\pi\gamma\ell_P^2\sqrt{(3/4)} = 8\pi \times 0.2375 \times (1.616 \times 10^{-35})^2 \times 0.8660$

Step 2:  $\Delta p = \hbar\nu/c$  (GW quantum momentum at frequency  $\nu$ )

Step 3: Robertson minimum:  $\Delta x = \hbar/(2\Delta p) = \hbar c/(2\hbar\nu) = c/(4\pi\nu)$

Step 4: Patch area:  $N_\odot \cdot A_j = \pi(\Delta x)^2 = \pi c^2/(16\pi^2\nu^2) = c^2/(16\pi\nu^2)$

Step 5:  $\nu^2 = c^2 / (16\pi \cdot N_\odot \cdot A_j)$

Step 6: Substitute  $A_j$ :  $\nu^2 = c^2 / (16\pi \cdot N_\odot \cdot 8\pi\gamma\ell_P^2\sqrt{(j(j+1))})$

Step 7:  $\nu^2 = c^2 / (128\pi^2\gamma\ell_P^2 \cdot \sqrt{(j(j+1))}) \cdot N_\odot = K_0 / (\sqrt{(j(j+1))}) \cdot N_\odot$

Step 8:  $K_0 = c^2 / (128\pi^2\gamma\ell_P^2) = 1.1467 \times 10^{84} \text{ Hz}^2$

The  $\sqrt{(j(j+1))}$  factor appears only in Step 7, not in  $K_0$  itself.

## Appendix C: $\sqrt{(j(j+1))}$ Placement Error Analysis

**Scenario 1 (self-canceling):** Absorb sf into  $K_0$  to get  $K_0' = 1.3241 \times 10^{84} \text{ Hz}^2$ , then omit sf at the frequency step. The absorption and omission cancel exactly.  $n$  is unchanged at 5.314.

**Scenario 2 (double application):** Absorb sf into  $K_0$  then also apply sf at the frequency step. The sf appears twice. This gives  $n = 5.21$ , outside the observed BBH mean by more than  $1\sigma$ . This is the error that produces a wrong result.

The correct procedure:  $K_0$  does not contain sf. The sf factor appears exactly once, in the denominator of the frequency equation.

## Appendix D: Change Log

### v18 vs v17:

1. Section 2.10 (new): Alpha Dual-Role analysis.
2. Section 5.2 (new): Alpha Gap mass-scaling law. Per-catalog table added.
3. Section 6.6 (new): Topological Factor of 2 unification. Five appearances enumerated.
4. Prediction 11 (new): IMBH regime prediction. BBH mergers at  $M \approx 1684 M_\odot$  will show gap  $\approx 1/\alpha$ .
5. Appendix E (new): Kerr Remnant Spin test. 264 events with posterior final\_spin values.
6. Abstract updated. Conclusions updated. Nine independent tests enumerated.

### v18.3 vs v18.2:

1. Abstract updated to summarise v18.3 results.
2. Section 5.2 extended: gap residual analysis. Empirical slope 0.922 vs algebraic 1.000 explained. Mass-ratio signal in gap residuals documented ( $r = -0.70$ ,  $p = 8 \times 10^{-40}$ ). Algebra closes to four decimal places:  $1.000 - 0.078 = 0.922$ .
3. Section 6.5 extended: precessing spin ( $\chi_p$ ) null result. Partial  $r = 0.20$  at full sample does not survive  $q > 0.70$  restriction ( $r \approx 0$ ,  $p = 0.99$ ). In-plane spin carries no independent face-activation signal.
4. Conclusions updated to reflect v18.3 results.
5. All empirical data, tables, formulas, and appendices carried forward from v18.2 unchanged.

## Appendix E: Kerr Remnant Spin Test

We tested whether replacing the Schwarzschild ISCO formula with the exact Kerr ISCO (Bardeen, Press & Teukolsky 1972) evaluated at the posterior remnant spin  $a_f$  reduces gap scatter or shifts the median gap toward  $1/\alpha$ .

Remnant spin distribution: median  $a_f = 0.6931$ , mean = 0.6859, std = 0.0995.

| Metric                                    | Schwarzschild | Kerr ( $\chi_{\text{eff}}$ ) | Kerr ( $a_f$ ) |
|-------------------------------------------|---------------|------------------------------|----------------|
| Median gap                                | 132.2328      | 132.1460                     | 131.1704       |
| Std of gap                                | 0.9147        | 0.9107                       | 0.9007         |
| Delta from $1/\alpha$                     | -4.8032       | -4.8900                      | -5.8656        |
| Kerr remnant shift                        | —             | -0.087 oct                   | -1.064 oct     |
| Partial $r(\chi_{\text{eff}}   \log_2 M)$ | -0.180        | -0.687                       | -0.468         |
| Partial $r(a_f   \log_2 M)$               | -0.846        | -0.971                       | -0.972         |

Key finding: the Kerr remnant correction shifts  $n_{\text{carrier}}$  by 1.064 octaves. The critical Kerr parameter at which this correction would equal the Robertson–Heisenberg 1.000-octave topological prediction exactly is  $a_f = 0.612$ . Observed median  $a_f = 0.693$  exceeds this critical value, overshooting the  $\chi(S^2) = 2$  exact equality by 0.064 octaves. Whether this surplus is fixed by the mass-ratio distribution of LIGO's BBH population or reflects a deeper constraint on remnant spins is a question for O5.

The Kerr remnant correction does not close the gap toward  $1/\alpha$  because  $a_f$  is almost completely determined by mass (partial  $r(a_f | \log_2 M) = -0.972$ ). The 4.80-octave systematic offset is a mass-scale effect, not a spin-correction effect.

$$n(C) = 3.561 + 3.506 \times C$$

*End of Preprint v18.3 — March 2026*